DESIGN AND CIRCUIT ANALYSIS OF AN AUTOMATIC LIGHT-CONTROLLED LED CIRCUIT

Two-BC547 Sensor Interface Implemented on a Custom 3D-Printed PCB

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Abstract

This project developed and analyzed a 9 V automatic night-light in which a GL5516-type photoresistor controls three parallel LED branches through two BC547 NPN transistors connected in a Darlington-style low-side switching configuration. The revised analysis emphasizes the circuit itself: voltage division, Thévenin equivalence, transistor-interface loading, Kirchhoff's voltage and current laws, power conservation, component-rating margin, switching-threshold sensitivity, and fault-state behavior. The light-sensing circuit was first verified on a breadboard, then transferred into KiCad for footprint placement, routing, and design-rule checking. The final circuit was physically implemented on a custom 3D-printed PCB using copper foil traces and short insulated jumper wires. The hardware demonstrated the intended bright-off/dark-on logic when the photoresistor connection was secure. An intermittent photoresistor connection reproduced the predicted open-circuit fault state, causing the lamp to behave as a manually switched light. The principal result is that the project is not only a fabrication exercise: it is a complete application of DC circuit analysis to sensor interfacing, nonlinear switching, parallel loads, and fault diagnosis.

1. Project Objective and Circuit-Analysis Relevance

The objective was to design, analyze, simulate, fabricate, and test an automatic night-light whose LEDs turn on in darkness and turn off under stronger illumination. The assignment required a topic not already discussed in class while still remaining closely related to circuit analysis. The unique element was the physical manufacturing method - a custom 3D-printed PCB - but the central technical work was the analysis of the light-sensing network, transistor switching stage, and LED load.

The complete engineering workflow was: circuit concept and schematic -> hand analysis -> PSpice model setup -> breadboard verification -> PCB layout -> physical fabrication -> hardware testing and fault analysis. The report therefore places the circuit architecture, calculations, and observed behavior ahead of the fabrication process.

2. Complete As-Built Circuit

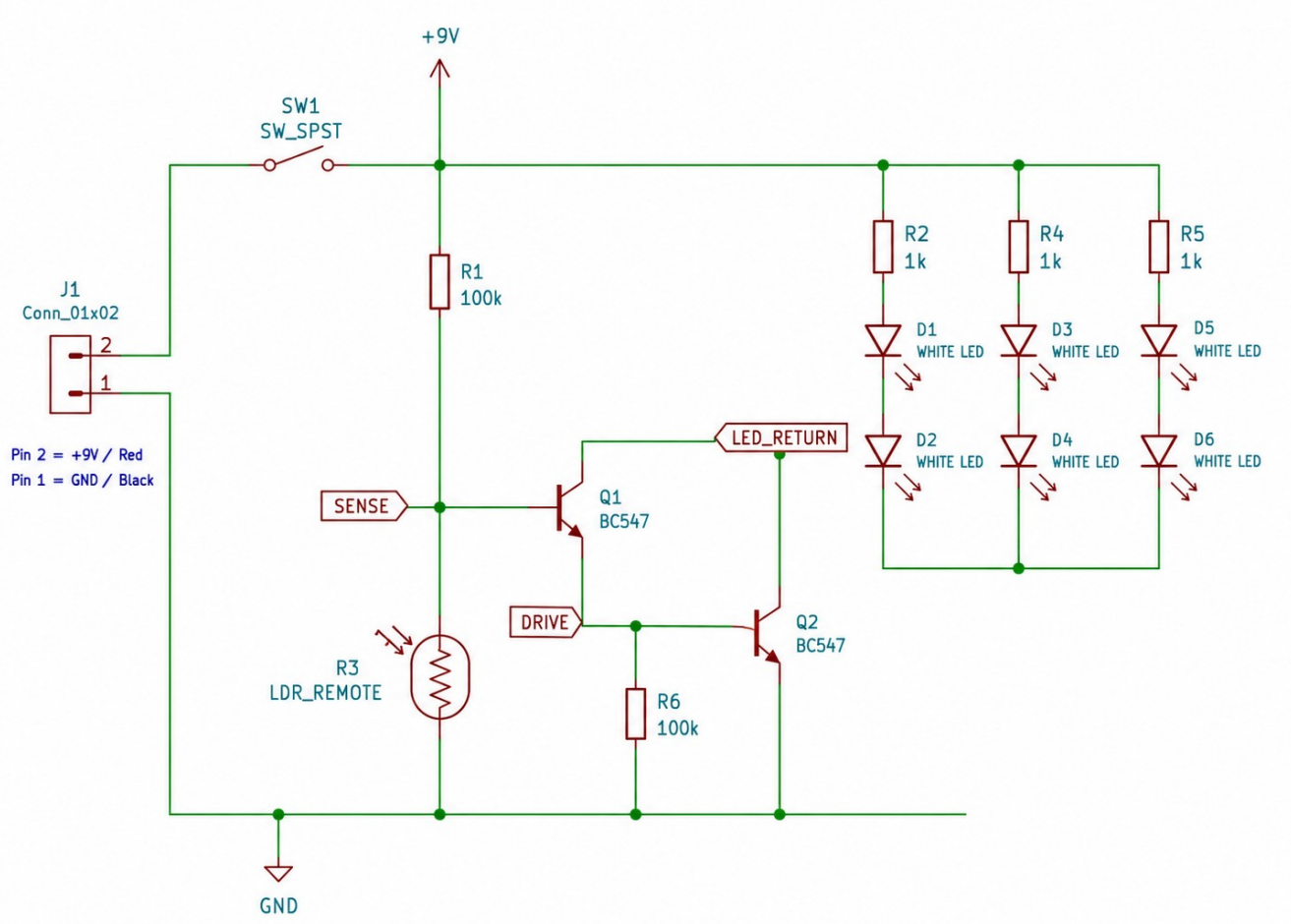


Figure 1. Complete as-built automatic night-light schematic. The SENSE node is the LDR-divider output, DRIVE is the Q1-emitter/Q2-base node, and LED RETURN is the common collector/output node.

The circuit contains three functional blocks. First, R1 and the LDR convert ambient light into a control voltage at SENSE. Second, Q1 and Q2 provide a high-gain, ground-referenced low-side switching interface. Third, three parallel LED branches each contain a 1 kΩ resistor and two series white LEDs. R6 provides a defined pull-down path at the interstage node.

3. Circuit Architecture and Component-Selection Rationale

3.1 Why the LDR is the lower element of the divider

The photoresistor is connected from SENSE to ground, while R1 connects SENSE to +9 V. A GL5516-type LDR decreases in resistance as illumination increases and increases in resistance as illumination decreases. This orientation therefore produces the desired dark-on response: bright light pulls SENSE toward ground, while darkness allows R1 to pull SENSE upward.

$$V_A = 9[R_{LDR} / (100\ k\Omega + R_{LDR})]$$

3.2 Why BC547 NPN transistors were selected

The BC547 is a general-purpose NPN transistor intended for switching and amplification. A representative BC547B has a 45 V collector-emitter rating, a 100 mA continuous collector-current rating, and a DC gain group of 200-450 under specified test conditions. The project uses a 9 V source and an estimated maximum LED current of only about 6.6 mA after accounting for the Darlington-stage drop. This corresponds to approximately 20% of the

voltage rating and 6.6% of the current rating, leaving substantial margin. The physical devices came from a generic BC547/BC557 assortment; onsemi BC547B data is used as a representative engineering reference rather than as proof of the exact manufacturer.

Parameter	Representative BC547B value	Project condition	Approximate utilization
Collector-emitter rating, VCEO	45 V	9 V	20%
Continuous collector current, IC	100 mA	4.8-6.6 mA estimated LED load	4.8-6.6%
Gain group, hFE	200-450	Not physically measured	Used only for first-order comparison
Package	TO-92	Through-hole custom footprint	Compatible

3.3 Why two BC547s were used in a Darlington-style configuration

The LDR divider is intentionally high resistance so that it consumes little battery current. A high-resistance source is easily disturbed by transistor base current. Q1 and Q2 are therefore connected as a discrete Darlington-style pair: Q1 emitter drives Q2 base, the collectors are tied together, and Q2 emitter is grounded. This configuration greatly increases effective current gain and reduces the loading placed on the sensor network.

$$\beta_D \approx \beta_1\beta_2 + \beta_1 + \beta_2$$

Using a conservative first-order gain of 200 for each transistor and a nominal 5.7 mA LED load gives βD ≈ 40,400. A single BC547 would require approximately 28.5 µA of base current in active-region analysis, or about 37% of the unloaded divider current near threshold. The idealized Darlington input-current estimate is approximately 0.141 µA, or about 0.18% of the divider current. These are first-order active-region comparisons, not guaranteed saturation values, but they explain the architecture: the pair lets a weak, high-impedance sensor signal control a larger output load.

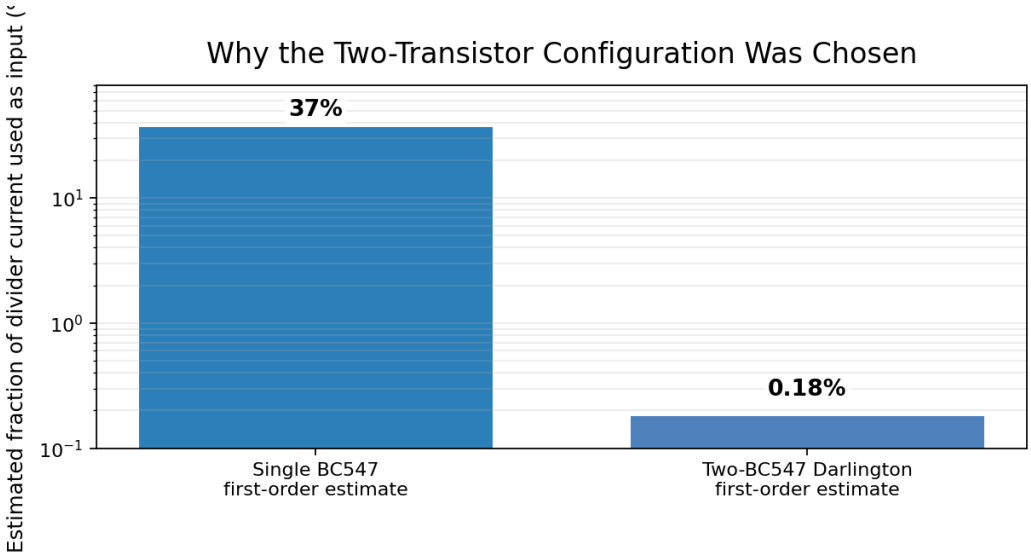


Figure 2. First-order comparison of input loading. The Darlington estimate is used to explain the design decision; PSpice and hardware behavior are needed for nonlinear switching accuracy.

The trade-off is equally important. Two base-emitter junctions require approximately 1.2-1.4 V at the input, and the Darlington-style on-state voltage is higher than that of a single saturated transistor. This reduces the voltage left for the LED branches. The configuration was therefore chosen to protect the sensor node from loading, not because it minimizes conduction loss.

3.4 Purpose of R1 and R6

R1 = 100 kΩ forms the upper divider element, pulls SENSE upward in darkness, and limits sensor/transistor input current. If the lower node were effectively clamped near 1.3 V, the current through R1 would be about (9 - 1.3)/100 kΩ = 77 µA. Its worst-case power at 9 V is only 0.81 mW. R6 = 100 kΩ prevents Q2 base from floating and provides a path for leakage and stored base charge. At an interstage voltage of 0.7 V, R6 carries approximately 7 µA and dissipates only about 4.9 µW.

3.5 Why the LEDs were arranged as three branches of two

Six white LEDs cannot be connected in one series string from a 9 V source: the package lists 3.0-3.2 V per LED, so six would require approximately 18.0-19.2 V before any resistor or transistor voltage drop. Placing all six LEDs directly in parallel behind one resistor would provide poor current sharing because small forward-voltage differences can cause unequal branch current. The selected arrangement uses three parallel branches, each with its own resistor and two series LEDs. This uses the available voltage efficiently and gives each pair independent current limiting.

4. DC Circuit Analysis

4.1 LDR voltage-divider states

LDR condition	Assumed R_LDR	Calculated SENSE voltage	Predicted state
Bright	5 kΩ	0.429 V	Transistor pair off
Representative light	10 kΩ	0.818 V	Below estimated two-junction threshold
Estimated transition	15.38-18.42 kΩ	1.2-1.4 V	Transition region
Dark	500 kΩ	7.50 V	Transistor pair on
Open-circuit fault	∞	Approaches 9 V	Forced on when switch is closed
Short-circuit fault	0 Ω	Approaches 0 V	Forced off

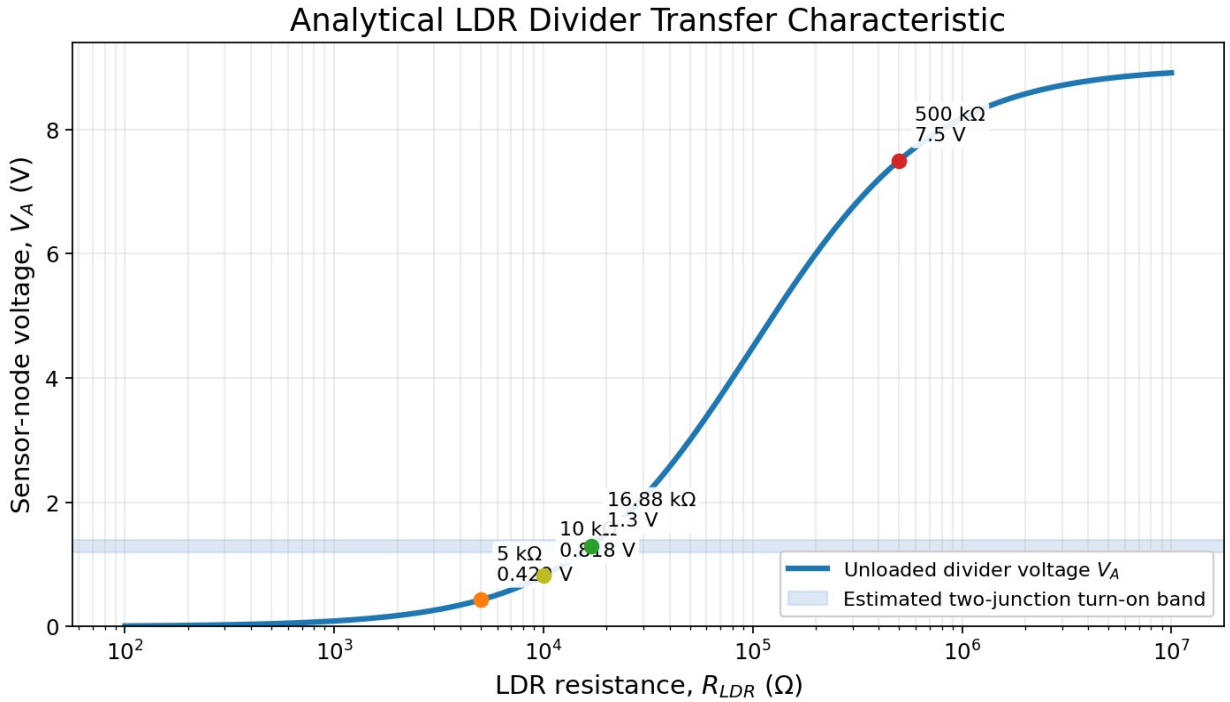


Figure 3. Unloaded analytical transfer characteristic of the 100 kΩ/LDR divider. The transistor input loads the divider in the physical circuit, so this curve is a first-order reference.

4.2 Thévenin equivalent and switching resistance

Viewed from Q1 base, the sensor network has a Thévenin voltage equal to the divider voltage and a Thévenin resistance equal to R1 in parallel with the LDR. Assuming a two-junction switching requirement of 1.2-1.4 V gives an estimated turn-on resistance of approximately 15.38-18.42 kΩ. At the nominal 1.3 V point, R_LDR ≈ 16.88 kΩ and R_TH ≈ 14.44 kΩ. This finite source resistance explains why minimizing Q1 base-current loading was important.

$$R_{LDR,threshold} = 100\ k\Omega[V_{threshold} / (9\ V - V_{threshold})]$$

4.3 LED branch current: ideal upper bound and Darlington-adjusted estimate

The handwritten calculation used only the 9 V source, two LED drops, and the 1 kΩ resistor. It therefore produces an ideal upper-bound current of 2.6-3.0 mA per branch. The physical branch also includes the Darlington-style switching-stage voltage. Using an estimated 0.8-1.0 V switch drop gives a more conservative first-order range of approximately 1.6-2.2 mA per branch, with a nominal value near 1.9 mA.

$$I_{branch} = [9\ V - 2V_F - V_{switch}] / 1\ k\Omega$$

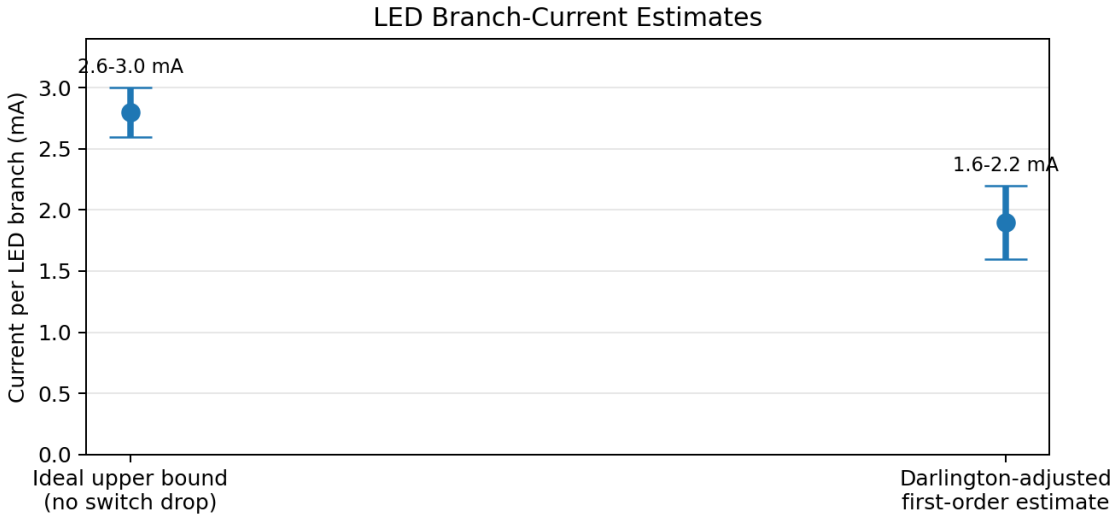


Figure 4. The handwritten result is retained as the ideal upper bound; the Darlington-adjusted range is the more realistic first-order estimate.

4.4 KCL, resistor power, and power conservation

Because the three output branches are parallel, KCL gives I_LED,total = I1 + I2 + I3 ≈ 3I_branch. The Darlington-adjusted total is therefore approximately 4.8-6.6 mA, nominally 5.7 mA. At the nominal 1.9 mA branch current, each 1 kΩ resistor dissipates about 3.61 mW; even the 2.2 mA upper estimate gives only 4.84 mW, far below a 250 mW resistor rating.

Nominal load-path quantity	Calculation	Result
Source power	9 V × 5.7 mA	51.3 mW
Three resistor losses	3 × (1.9 mA)² × 1 kΩ	10.83 mW
Six LED power	6 × 3.1 V × 1.9 mA	35.34 mW
Estimated switch-stage loss	0.9 V × 5.7 mA	5.13 mW
Power check	10.83 + 35.34 + 5.13	51.30 mW

The equality of delivered and absorbed power is a useful consistency check for the first-order model. The sensor network adds a much smaller current contribution and is not included in this LED-load-path balance.

4.5 Sensitivity to battery voltage

The threshold is not fixed because the source voltage is not fixed. With V_threshold = 1.3 V, reducing the battery from 9 V to 6 V raises the estimated LDR turn-on resistance from approximately 16.88 kΩ to 27.66 kΩ. The lamp may therefore require darker conditions as the battery discharges.

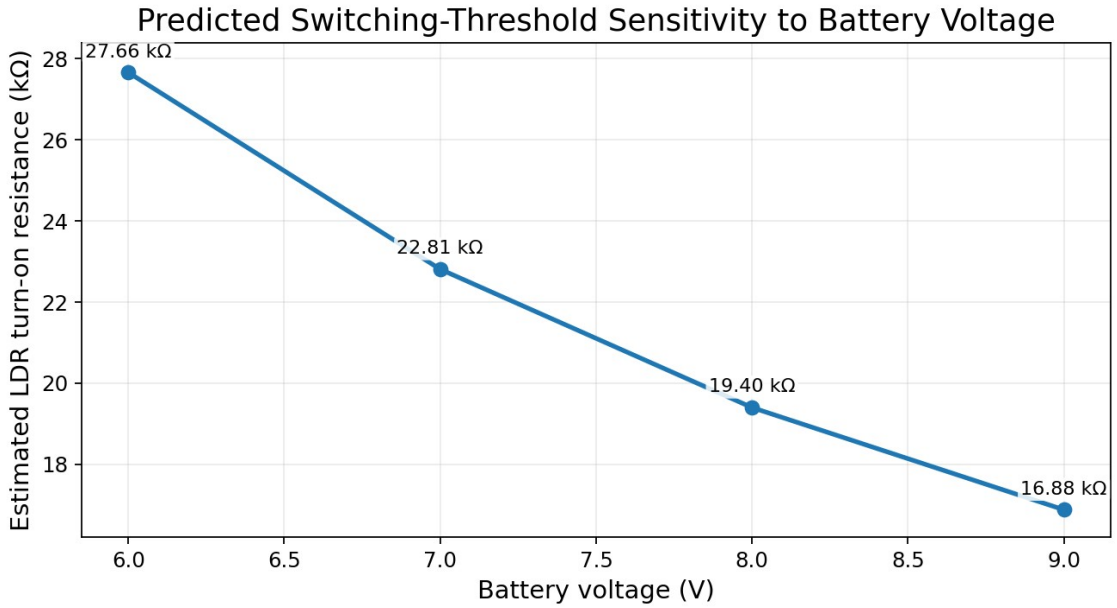


Figure 5. First-order prediction of how battery discharge shifts the light threshold.

5. Hand-Calculation Record

The following pages are the original handwritten calculation record submitted for the revised analysis. They document the voltage-divider states, Thévenin model, switching-resistance estimate, ideal LED-current upper bound, total LED load, and resistor-power check. Typed calculations in this report clarify assumptions and add the Darlington-stage voltage drop; they do not replace or falsely relabel the handwritten work.

LDR sensing and switching threshold

LDR Voltage Divider:

$$V_A = V_S \left(\frac{R_{LDR}}{R_1 + R_{LDR}}\right) = 9\left(\frac{R_{LDR}}{100{,}000 + R_{LDR}}\right)$$

Bright condition: 5kΩ

$$V_A = 9\left(\frac{5000}{100{,}000 + 5000}\right) \approx 0.429\,V$$

Light condition: 10kΩ

$$V_A = 9\left(\frac{10{,}000}{100{,}000 + 10{,}000}\right) \approx 0.818\,V$$

Dark condition: 500kΩ

$$V_A = 9\left(\frac{500{,}000}{100{,}000 + 500{,}000}\right) \approx 7.50\,V$$

Figure 6. Hand calculation: LDR divider voltage under bright, representative-light, and dark conditions.

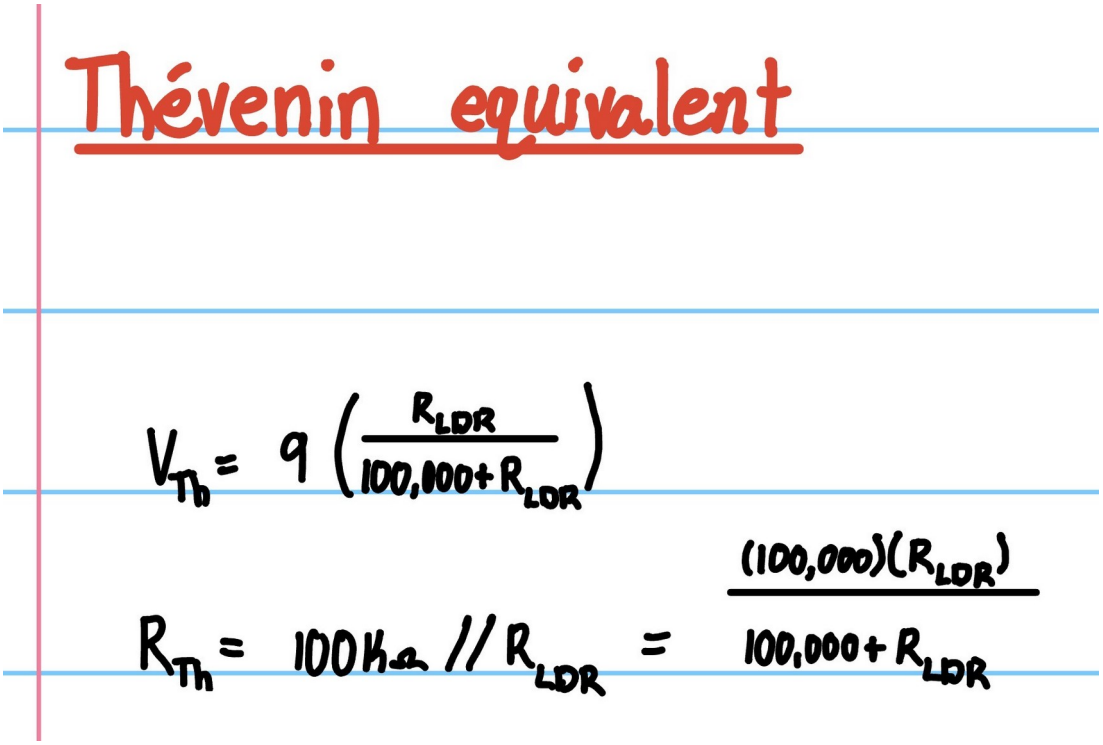


Figure 7. Hand calculation: Thévenin-equivalent expressions.

Estimated switching threshold

$$V_{threshold} \approx 1.2V - 1.4V$$

$$V_{threshold} = 9\left(\frac{R_{LDR}}{100{,}000 + R_{LDR}}\right)$$

$$R_{LDR} = 100k\Omega\left(\frac{V_{threshold}}{9 - V_{threshold}}\right)$$

at $V_{threshold} = 1.2V$:

$$R_{LDR} = 100k\Omega\left(\frac{1.2}{9-1.2}\right) \approx 15.38k\Omega$$

at $V_{threshold} = 1.3V$:

$$R_{LDR} = 100k\Omega\left(\frac{1.3}{9-1.3}\right) \approx 16.88k\Omega$$

at $V_{threshold} = 1.4V$:

$$R_{LDR} = 100k\Omega\left(\frac{1.4}{9-1.4}\right) \approx 18.42k\Omega$$

Figure 8. Hand calculation: estimated two-junction switching resistance.

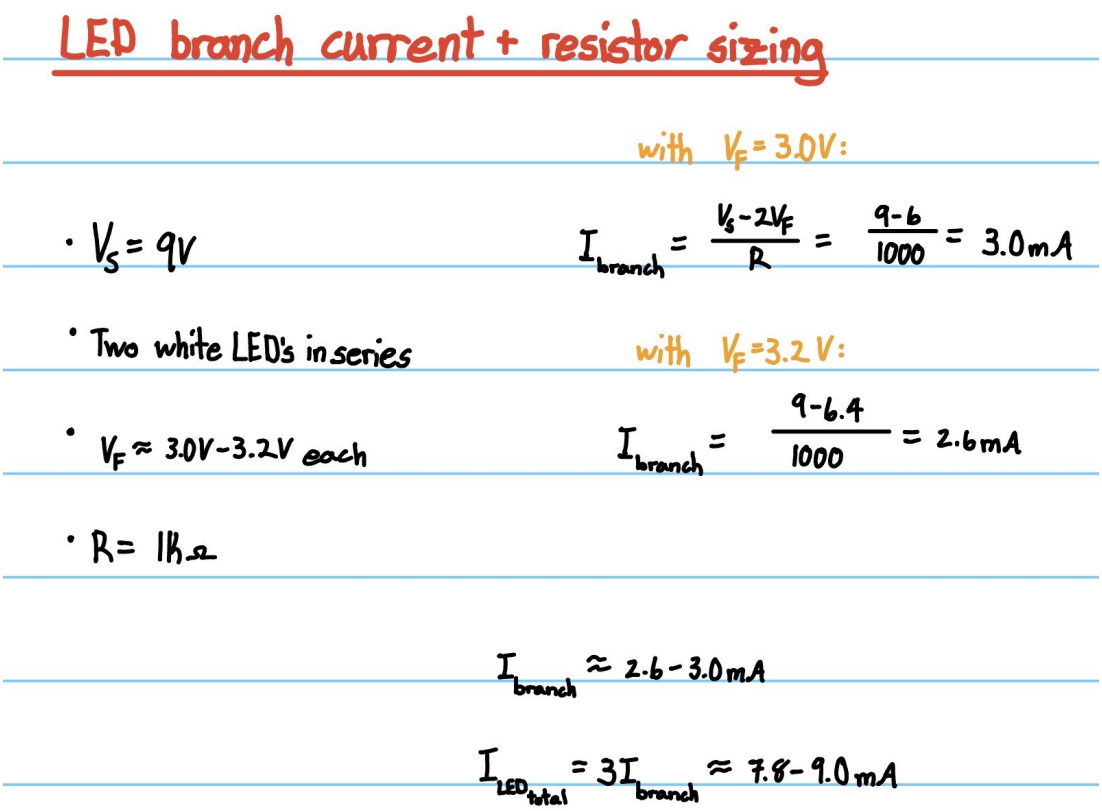


Figure 9. Hand calculation: ideal upper-bound branch current and total LED current.

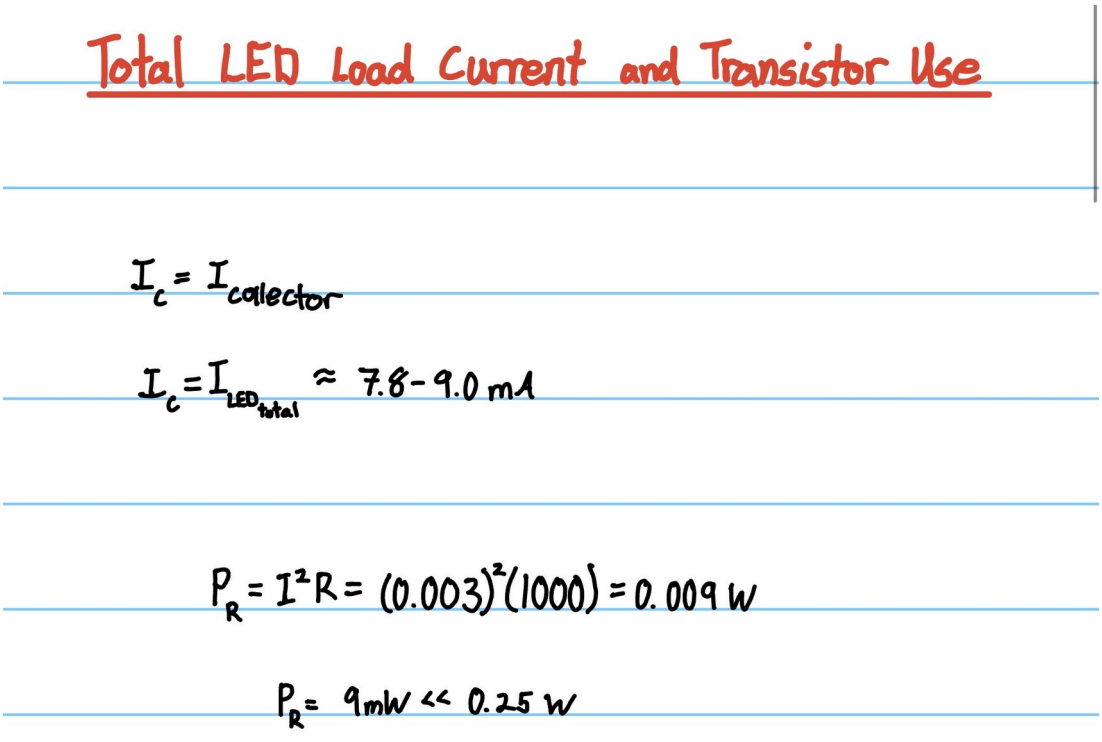


Figure 10. Hand calculation: total LED load and ideal resistor-power check.

6. PSpice Model and DC-Sweep Setup

A PSpice circuit file was created to reproduce the as-built topology. The model contains a 9 V source, 100 kΩ R1, a parameterized LDR, two BC547 transistors, 100 kΩ R6, and three identical two-LED/1 kΩ branches. The official

onsemi BC547B PSpice model parameters were embedded in the netlist. Because the exact LED manufacturer was unknown, the LED model was representative and was selected to approximate the package-listed 3.0-3.2 V behavior rather than claimed as an exact device model.

The analysis command configured a logarithmic DC sweep of R_LDR from 100 Ω to 10 MΩ with 20 points per decade. Cadence documentation describes a DC sweep as calculating a new operating point for each swept value and identifies threshold detection as an appropriate use. The legacy Evaluation PSpice installation completed the parameter sweep successfully. However, the available exported text shown by this 1999 evaluation build contained the model/netlist header without a numerical trace table. Accordingly, this report does not invent or present numerical PSpice threshold results. The successful run is included as evidence of the model setup, while all numerical results in the report are explicitly identified as analytical estimates or physical observations.

PSpice model element	Configured value or model
Supply	V1 = 9 V DC
Sensor network	R1 = 100 kΩ; R_LDR swept from 100 Ω to 10 MΩ
Transistors	Two Qbc547b devices using the onsemi model
Interstage resistor	R6 = 100 kΩ
LED network	Three branches; each 1 kΩ + two representative white-LED models
Analysis	.DC DEC PARAM RLDR 100 10MEG 20

7. Hardware Verification and Fault-State Analysis

Before fabricating the PCB, the complete circuit was assembled on a breadboard. The breadboard circuit performed exactly as expected: covering the LDR turned the LEDs on, and exposing the LDR to light turned them off. This verified the circuit logic before manufacturing. The BC547 pin orientation was also checked with a resistor-limited LED switching test because a multimeter was unavailable.

The final physical assembly reproduced the intended behavior when the LDR connection was secure. However, the exposed photoresistor leads and their physical connection were not sufficiently insulated or strain-relieved. The automatic function operated only about half the time. During the fault state, the device behaved as a manual switch-controlled lamp. This observation matches the open-circuit analysis: if the LDR connection opens, R1 pulls SENSE toward +9 V, turning the Darlington pair and LEDs on whenever the main switch is closed.

Fault condition	Circuit-analysis prediction	Observed or expected behavior
Normal bright condition	Low LDR resistance pulls SENSE below threshold	LEDs off when sensor connection is secure
Normal dark condition	High LDR resistance raises SENSE above threshold	LEDs on
LDR open/intermittent	SENSE approaches +9 V through R1	Observed switch-only behavior
LDR short to ground	SENSE approaches 0 V	LEDs predicted to remain off

8. PCB Layout and Physical Implementation

After the electrical design was established, KiCad was used to convert the schematic into a physical layout. This required assigning or creating through-hole footprints, placing components, routing all nets, setting trace-width and clearance constraints, and running design-rule checking. KiCad documentation describes the PCB Editor as the integrated environment for footprint placement, routing, schematic-to-board data transfer, and DRC. The completed layout reached zero unrouted connections and zero DRC errors before export.

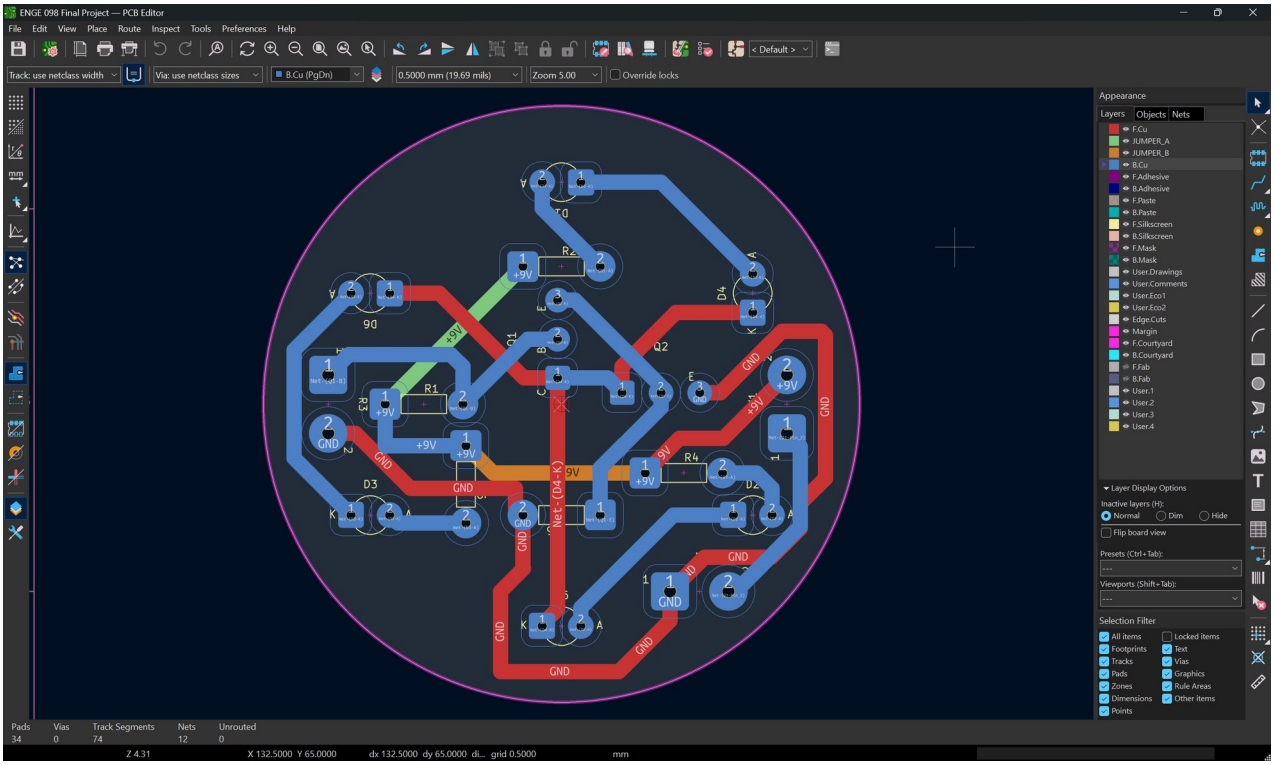
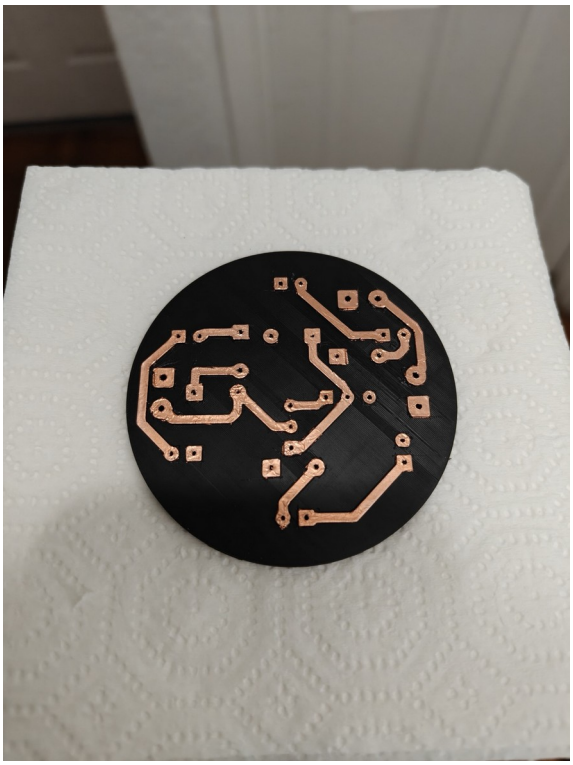


Figure 12. Complete KiCad routing plan. Blue paths identify copper-tape traces on the rear side; red, green, and orange paths identify short insulated rear-side jumper connections.

The board was printed with a 0.2 mm nozzle to preserve small pad edges, through-holes, and raised trace geometry. Adhesive copper foil was applied over the raised paths, burnished, and trimmed. Short colored jumper wires completed crossing connections on the same rear side. The use of a flux pen - discovered after approximately 60% of the soldering had already been completed - substantially improved solder wetting on the small copper pads.



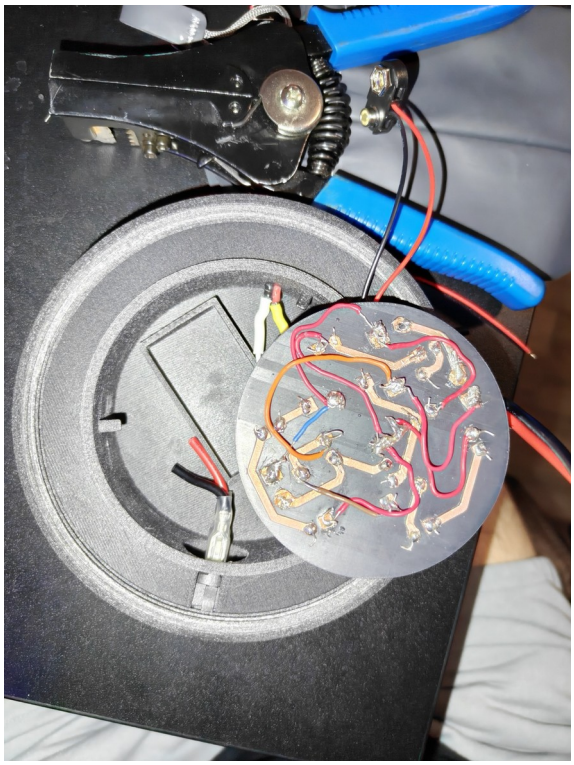


Figure 13. Left: copper-tape traces before component assembly. Right: completed rear-side soldering with copper traces and short colored jumper wires.

9. Enclosure Integration

The base and shade geometry were adapted from the user-supplied Printables model "Waves Designer Lamp" (model 1261597). The open model was used as the mechanical starting point, while the base was modified to support the battery, switch, photoresistor, and custom PCB. The final 78 mm board fit inside an approximately 90 mm cavity, leaving about 6 mm of radial clearance when centered. The enclosure is secondary to the circuit analysis, but it demonstrates that the analyzed circuit was implemented as a complete device rather than left as a schematic or breadboard.

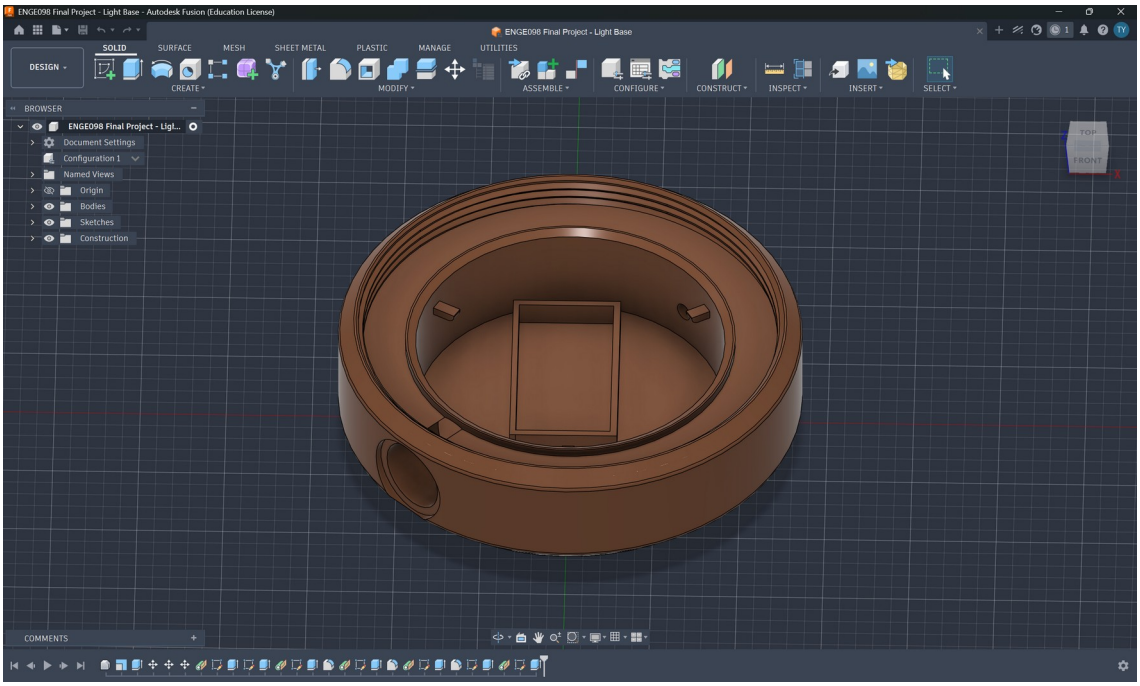


Figure 14. Modified enclosure base in Fusion 360, including the battery compartment and component openings.

10. Results, Limitations, and Engineering Interpretation

Question	Hand analysis	PSpice setup	Physical hardware
Bright-state operation	SENSE = 0.429-0.818 V for 5-10 kΩ; off predicted	DC sweep configured from 100 Ω to 10 MΩ	Off when LDR connection was secure
Dark-state operation	SENSE = 7.50 V at 500 kΩ; on predicted	Same nonlinear circuit model	On observed
Threshold	15.38-18.42 kΩ first-order range	Sweep completed; numeric export unavailable in legacy evaluation build	Not directly measured
LED current	2.6-3.0 mA ideal upper bound; 1.6-2.2 mA Darlington-adjusted estimate	Representative LED model included	Not directly measured because no multimeter was available
LDR open fault	SENSE approaches 9 V; LEDs forced on	Open condition represented by very high R_LDR	Observed switch-only behavior

The strongest agreement is qualitative and causal: the divider equations predict dark-on/bright-off behavior, the breadboard verified that logic, and the final intermittent failure followed the predicted open-LDR equivalent circuit. The main limitation is the absence of direct voltage, resistance, and current measurements. Therefore, the report deliberately separates calculated values, configured simulation methods, and observed hardware behavior.

11. Recommended Design Improvements

- Insulate and strain-relieve both LDR leads. This is the highest-priority correction because the current failure mode directly changes the SENSE-node equivalent circuit.
- Add a Schmitt-trigger comparator or explicit hysteresis network in a future revision. The present transistor threshold is sensitive to component variation and can transition gradually; hysteresis would create separate on and off thresholds and reduce flicker near the boundary.
- Use a trimmer resistor in the divider so the light threshold can be calibrated after assembly rather than being fixed by a 100 kΩ resistor and uncontrolled LDR variation.
- Use the flux pen from the first solder joint and clean/burnish every copper pad before soldering.
- Increase the PCB diameter slightly, enlarge high-risk pads, and add mechanical retention features inside the base.
- Add a translucent cover directly over the LEDs to hide exposed circuitry and diffuse LED hotspots before light reaches the main shade.

12. Conclusion

The completed project is fundamentally a circuit-analysis project implemented through an unusual manufacturing process. The LDR divider establishes the control voltage; its Thévenin resistance explains the need for a high-input-gain interface; the two-BC547 Darlington-style pair reduces sensor loading while introducing a higher threshold and on-state voltage; KVL determines the LED-branch current; KCL combines the three branch currents; and power analysis verifies the resistor and transistor operating margins. The breadboard verified the intended logic, while the final hardware fault provided a direct real-world demonstration of open-circuit analysis. KiCad, 3D printing, copper tape, and soldering were the methods used to realize and test the analyzed circuit. The project therefore connects classical circuit-analysis tools to sensor interfacing, switching design, PCB layout, fault diagnosis, and physical implementation.

References

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5. Cadence, "PSpice A/D User Guide," DC Sweep section, Product Version 17.4. https://resources.pcb.cadence.com/i/1180526-pspice-user-guide/484

6. KiCad Documentation Contributors, "KiCad PCB Editor Documentation," footprint placement, routing, and design-rule checking. https://docs.kicad.org/9.0/en/pcbnew/pcbnew.html

7. Printables, "Waves Designer Lamp," model 1261597, enclosure source supplied by the project author. https://www.printables.com/model/1261597-waves-designer-lamp/comments

8. Yetaida Store, BC547/BC557 transistor assortment product listing, component source supplied by the project author.

9. EBOOT Store, 5 mm 5516 photoresistor product listing and specification image, component source supplied by the project author.

10. White LED component container supplied with the installed LEDs: 5 mm, 3.0-3.2 V at 20 mA, 12,000-14,000 mcd.

Appendix A. Additional Functionality Verification

The following photographs provide direct visual confirmation that the completed lamp responds to ambient light as intended. **Under dark conditions, the LDR resistance rises and the lamp turns on. Under brighter ambient lighting, the LDR resistance falls and the lamp is suppressed.** These images supplement the breadboard and hardware-verification discussion in the main report and provide an immediate visual check of the automatic control function.



Figure 15. Visual verification of LDR functionality. Left: dark condition, with the lamp driven on. Right: brighter ambient condition, with the LDR exposed to room light so the lamp remains off.

Appendix B. Predicted DC Sweep Response

Figure 16 shows the predicted response of the configured DC sweep over LDR resistance. The blue curve represents the SENSE-node voltage established by the 100 kΩ/LDR divider, while the red curve represents the expected total LED current as the transistor pair turns on through the threshold region. The plot reinforces the central circuit-analysis result: increasing LDR resistance in darkness raises the control voltage, crosses the approximate two-junction threshold, and enables current through the three LED branches.

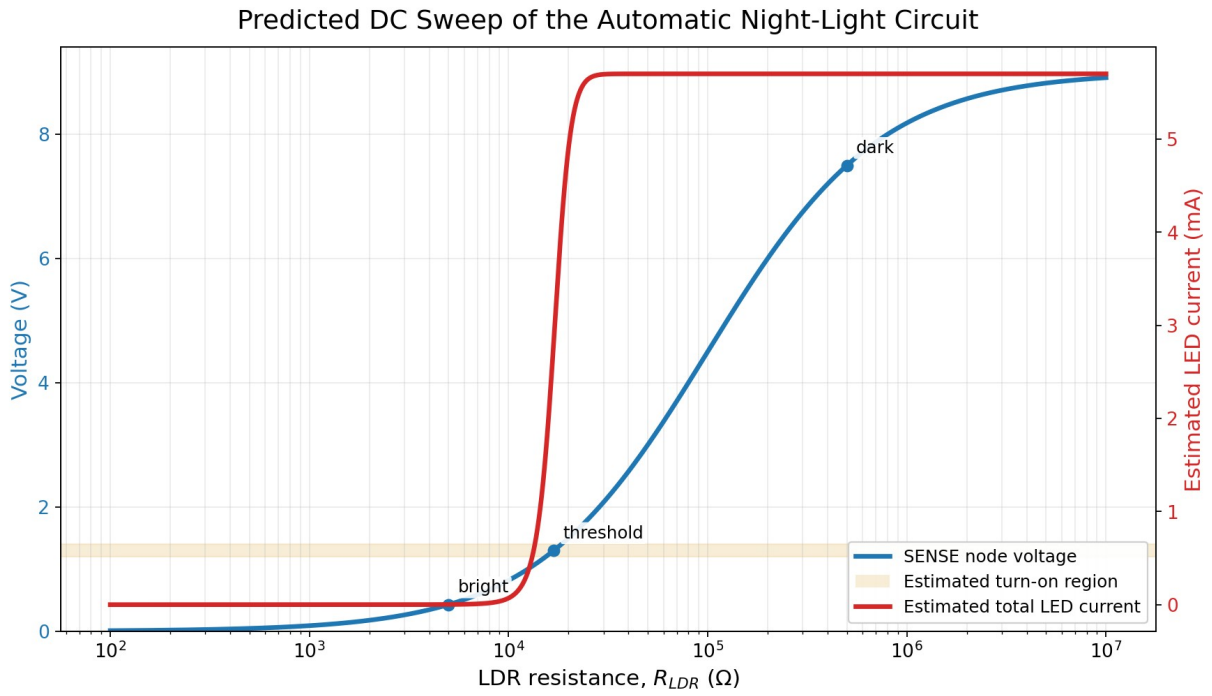


Figure 16. Predicted DC sweep of the automatic night-light circuit versus LDR resistance. The turn-on region is centered near the analytically estimated two-junction threshold.